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Associations of Type 2 Diabetes with risk of overall and site-specific cancers in a cohort of predominantly low-income racially diverse populations

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Abstract

Background: The prevalence of type 2 diabetes (T2D) is higher in Black than white Americans, and individuals with T2D have an increased cancer risk. We investigated the association of

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Concept and design: Zheng and Pradhan.

Acquisition, analysis, or interpretation of data: Pradhan, Zheng, and Wen.

Drafting of the manuscript: Pradhan, Zheng.

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T2D with the risk of all cancer combined and 21 site-specific cancers, among predominantly low-income participants who experienced a disproportionately high risk of both T2D and cancer.

Materials and Methods: The study included 76,121 participants (mean age: 52.0 years; 67.2% Black) from the Southern Community Cohort Study (SCCS). T2D was ascertained at the baseline survey. Incident cancer was ascertained via linkage to state cancer registries. Cox proportional hazard models were used to estimate the associations between T2D and cancer after adjusting for confounders.

Results: Among participants, 21.2% (N=16,137) had baseline T2D, and 9.7% (N=7,376) were diagnosed with incident cancer. Compared to individuals without T2D, individuals with T2D had a significantly elevated risk of all cancer combined (HR: 1.07; 95% CI: 1.01–1.13) and several site-specific cancers, including cancers of the stomach, colorectum, pancreas, liver/intrahepatic bile duct, kidney, and renal pelvis as well as leukemia. The significant association for most cancers was largely observed within 15 years after T2D diagnosis, except for cancer of the pancreas and liver/intrahepatic bile duct, for which elevated risks remain statistically significant 15 to 30 years after T2D diagnosis

Conclusion: T2D was associated with the risk of overall and certain site-specific cancers in this predominant low-income population.

Impact: Preventive measures to reduce the burden from T2D could help reduce the risk of overall cancer and several site-specific cancers.

Keywords

diabetes; cancer; incidence; racial disparities; epidemiology

INTRODUCTION

The prevalence of type 2 diabetes (T2D) in the United States is 11.3%, with 37.3 million living with diagnosed T2D and an additional 8.5 million who remain undiagnosed(1). Among individuals who are ≥ 65 years, the prevalence of cancer is 17.3% for those with T2D compared to 16.4% for those without T2D; while among individuals aged 30–64 years, the prevalence of cancer is 7.6% in those with T2D compared to 5.3% for those without(2). Both T2D and cancer share certain modifiable risk factors, such as obesity, lack of physical activity, unhealthy diet, excessive alcohol consumption, and tobacco smoking(3). Previous research conducted among European and Asian populations suggests that individuals with T2D are at an increased risk of developing certain types of cancer(4,5). This could be due to the shared lifestyle risk factors and underlying biological mechanisms(6–10).

Additionally, race is also a factor that could disproportionately affect the T2D-cancer association(3,11). The prevalence of T2D is much higher in Black individuals with cancer compared to White individuals with cancer(12). Since the risk of having T2D is higher for Black individuals than for White individuals, and previous research has suggested that individual poverty could increase the risk of having T2D, there may be an increased risk among Black individuals who have T2D and also experience poverty(6,13–16).

Currently, limited prospective studies have been conducted to evaluate the associations between T2D and cancer incidence in underserved populations who bear a high prevalence of T2D. To bridge the gap in epidemiologic evidence, we conducted a prospective cohort study to investigate the association of T2D with overall cancer as well as 21 site-specific cancers among predominantly low-income Black and White participants in the Southern Community Cohort Study (SCCS).

MATERIALS AND METHODS

Study Population

The details of the SCCS are described elsewhere(17,18). In brief, it is a prospective cohort study of nearly 86,000 adults recruited across 12 states in the southeastern United States (Alabama, Arkansas, Florida, Georgia, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Virginia, and West Virginia)(19); approximately 86% of them were recruited from community health centers that provide basic health care to low-income populations, and the remaining 14% were recruited from the general population via a mailed survey or a telephone interview. The cohort has a high representation of Black individuals (~65%) and women (~65%), and more than half of the participants have a household income of less than \$15,000/year(20). The baseline prevalence of T2D in this cohort is approximately 20%. Demographic, socioeconomic, lifestyle, and anthropometric data, as well as personal medical history, were ascertained at cohort enrollment via standardized computer-assisted in-person interviews for community health center participants and self-administered mailed questionnaires for persons recruited from the general population. Enrollment in the cohort began in March 2002 and concluded in September 2009.

Assessment of T2D and Cancer

T2D was ascertained at the baseline survey by asking study participants whether they had been previously diagnosed by a physician with the condition. Incident cancer was ascertained via linkage to state cancer registries and was classified according to the primary site and histology categories defined by the Surveillance, Epidemiology, and End Results (SEER) Program (SEER Data Reporting Tools [cited 2023 January 23] Available from: https://seer.cancer.gov/siterecode/icdo3_d01272003/index.html). Cancers were classified using the International Classification of Diseases (ICD)-O version 3 (Southern Community Cohort Study ICD-O Codes [cited 2023 January 23] Available from: https://www.southerncommunitystudy.org/uploads/5/2/7/5/52750661/sccs_incident_cancer_11_5_2021.pdf). Only cancer sites with 50 incident cases or more were included in the analysis. The vital status of study participants was determined via linkage with the National Death Index.

Statistical Analysis

We excluded those participants with missing information at the baseline survey for a prior diagnosis of T2D (N=1,513; 1.8%) and cancer (N=2,684; 3.2%) or race (N=800; 0.9%), not mutually exclusive. Then we further excluded those participants who reported at baseline with cancer except for non-melanoma skin cancer (N=6,684; 7.9%). After exclusions,

76,121 participants remained for the current analysis. Among those, there were 16,137 individuals with T2D at baseline.

Follow-up time started at the age of the baseline survey and ended at the time of cancer diagnosis, death, or the last date of linkage to identify incident cancer cases, whichever came first. The date of the last cancer registry linkage differed by states, ranging from 2016 to 2020, and mostly in 2018–2019. For this analysis, we used Cox proportional hazards models to evaluate the association of T2D with cancer risk. We used age as the time scale in the Cox regression model. All models were adjusted for sex (male or female), race (Black, White, or other), education (less than high school, completed high school or general education development or other training, some college or junior college, graduated from college or further education), annual household income (<\$15,000, \$15,000–\$24,999, \$25,000–\$49,000, \$50,000–\$99,999, \$100,000+), enrollment source (community health center in-person interview, general population mailed survey, or general population telephone interview), smoking status (current, former, or never), pack-years of smoking, and body-mass index (<18.5, 18.5–24.99, 25–29.99, 30–34.99, ≥ 35.0 kg/m²). Alcohol intake was not adjusted in the analysis as it was not a confounder as evaluated using the directed acyclic graph (DAG) analysis. Furthermore, there was no statistical difference among individuals who had diabetes or did not have diabetes regarding alcohol intake (P -value = 0.2679). Additionally, we conducted stratified analyses by race (white and black), years between the diagnosis of type 2 diabetes and the baseline survey (<10 years, 10–19.99 years, and ≥ 10 years), and years since diabetes diagnosis (<15 years, 15–29.99 years, and ≥ 30 years). The years since the diabetes diagnosis were treated as a time-varying variable in the Cox regression models. We only analyzed cancers with more than 50 cases to have stable estimates. The Cox proportional hazards assumption was formally assessed using graphical visuals and statistical tests based on the scaled Schoenfeld residuals, and this assumption generally held for the associations evaluated in this study.

Analyses were further stratified by gender, race, smoking status, BMI, and follow-up time (< 5 years and ≥ 5 years). For the analysis stratified by follow-up time, person-years for <5 years of follow-up were calculated from all eligible participants for the first five years of follow-up, while person-years for ≥ 5 years of follow-up were calculated excluding the first 5-year follow-up for those with ≥ 5 years of follow-up. Interactions were assessed with likelihood ratio tests to compare the models with and without the addition of cross-product terms.

The missing values of these covariates (less than < 5% missing for all covariates in general) were first imputed using the multivariate imputation by chained equations to preserve the data(21). Of these, the missing categorical covariates were household income (N=2,618; 3.1%); education (N=1,848; 2.2%); and smoking status (N=2,095; 2.5%). The missing continuous covariate was BMI (N=2,393; 2.8%). All P values presented were two-sided, with statistical significance set at 0.05.

Data Availability Statement

The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request.

RESULTS

The descriptive and demographic characteristics of the study population for the overall study population and by T2D status are presented in Table 1. Among those with T2D, the mean baseline age was 52.0 years, 41.4% were males, 67.2% were Black adults, and 64.0% were former or current smokers. Among those without T2D, the mean baseline age was 54.8 years, 35.6% were males, 69.5% were Black adults, and 58.6% were either former or current smokers.

Figure 1 portrays the associations between T2D and the risk of overall cancer or site-specific cancers. Compared to individuals without T2D, individuals with T2D had an elevated risk of all cancers combined (HR: 1.07; 95% CI: 1.01–1.13). Individuals with T2D also had an elevated risk of certain site-specific cancers, such as stomach, colorectal, pancreatic, liver, and intrahepatic bile duct, and kidney and renal pelvis cancers as well as leukemia, and a reduced risk of prostate cancers. Analyses were also performed after excluding those individuals with < 5 years of follow-up time (Supplemental Table 1).

Table 2 shows the association between T2D and cancer risk stratified by race. White individuals with T2D had a significantly elevated risk of all cancers combined and several site-specific cancers, including colorectal and liver and intrahepatic bile duct cancers. Similar to White individuals, Black individuals with T2D also had a significantly elevated risk of any cancer, as well as liver and intrahepatic bile duct cancers. In addition, Black individuals with T2D also had an elevated risk for stomach, pancreatic, leukemia, and kidney and renal pelvis cancers. However, the racial difference in the strength of the T2D-cancer association was only statistically significant for all cancers combined and uterine body and liver and intrahepatic bile duct cancers (P for heterogeneity < 0.05).

We also assessed cancer incidence stratified by years between the baseline survey and diagnosis of T2D (Table 3). In order to provide stable risk estimates, we included in this analysis only cancer types with >50 cases diagnosed with T2D. Results show that individuals with T2D have a slightly decreasing risk of cancer as the years increase, where those with T2D for ≥ 20 years have an HR of 0.96 (95% CI: 0.83–1.09). Among individuals with T2D, there is a significantly decreasing risk of lung cancer as the years increase..

We also assessed associations of T2D diagnosis with cancer risk according to years since T2D diagnosis using time-varying Cox regression models (Table 4). Similarly, we included in this analysis only cancer types with >50 cases diagnosed with T2D. Results show that individuals with T2D have an increasing risk of all cancers combined within 10 years after T2D diagnosis, with an HR of 1.10 (95% CI: 1.02–1.17). The risk begins to decrease with an increasing duration since T2D diagnosis. A similar pattern of the associations was found for virtually all site-specific cancers evaluated in Table 4, except for cancers of the pancreas and liver/intrahepatic bile duct, for which elevated risks remain statistically significant even 15 to 30 years after T2D diagnosis.

We also performed stratified analyses by sex (Supplemental Table 2), smoking status (Supplemental Table 3), BMI (Supplemental Table 4), and by both race and income (Supplemental Table 5 and Supplemental Table 6). Overall, no apparent modifying effect

was observed, and the tests for interactions were generally not significant after adjusting for multiple comparisons.

DISCUSSION

In this large prospective cohort study, we found an increased risk of overall cancer in association with T2D. Those with T2D had an elevated risk of certain site-specific cancers, such as leukemia as well as stomach, colorectal, pancreatic, liver and intrahepatic bile duct, and kidney and renal pelvis cancers; and a reduced risk of prostate cancer. This study provides supporting evidence that some findings observed previously in other populations regarding the association between T2D and cancer risk can be extended to the low-income and Black population, while some other findings are unique.

Limited studies have assessed the associations of T2D with the overall risk of developing cancer in Black populations. However, there are studies analyzing this association in other cohorts. Results from a study by Bjornsdottir et al. using data from a Swedish cohort found that the risk of any cancer among T2D patients had an HR of 1.19 (95% CI: 1.09–1.12). Another study by Inoue et al. using data from a Japanese cohort found similar results with an HR of 1.27 95% CI: 1.14–1.42(4,5). Data combined from previous observational studies and meta-analyses have shown that T2D is positively associated with increased risk of all cancers and several site-specific cancers, including pancreatic, liver, kidney, bladder, breast cancer, colorectal, and endometrial, while being negatively associated with risk of prostate cancer(2,22). We found a weak positive association of T2D with all cancers combined (HR: 1.07; 95% CI: 1.01–1.13). The reasons for this inconsistency are unclear. Our findings for site-specific cancers are similar to those reported previously in other populations.

With a few exceptions, the associations between T2D and cancer risk were found to be similar between Black and White individuals in this study. The association of T2D with the risk of overall cancer and cancer of the liver and intrahepatic bile duct appeared to be stronger in White than Black study participants (P for heterogeneity < 0.05). The reasons for this finding are unclear. It is possible that other cancer risk factors are more prevalent in Black than white participants in this study, which partially masked the positive association of T2D with overall cancer risk in Black participants. Surprisingly, a prior T2D diagnosis showed a positive association with uterine body cancer in White participants but an inverse association in Black participants (P for heterogeneity < 0.05). Again, the reasons for this difference are unclear. However, we could not rule out possible chance findings due to multiple comparisons.

Overall, we found a weaker association between T2D and cancer risk in this study than previously reported. It is possible that the risk of cancer in this low-income population, including those without prior diabetes, may have masked some of the association signals with T2D in our study, resulting in a weaker association observed in this study compared with some previous studies conducted in other populations. Nevertheless, our study did provide additional evidence for a positive association of T2D with the risk of certain cancers, which can help understand implications from T2D on cancer in low-income and minority populations whom previous analyses may not well represent.

Certain mechanisms may explain the increased risk of cancer among diabetic individuals. These plausible mechanisms affecting cancer etiology and accelerating this progression could be hyperglycemia, hyperinsulinemia, and inflammation(3,23). Hyperglycemia could promote the proliferation of cancer cells and cell growth(3). Similarly, previous studies have shown that hyperinsulinemia could increase the expression of insulin growth factor-1 and activate the receptor, leading to further cancer cell stimulation and growth(24). For example, individuals with diabetes could be at an increased risk of pancreatic cancer due to hyperinsulinemia, glucose toxicity, excess iron in the circulation, exogenous insulin therapy, and impaired B-cell function(23,25–27). For the SCCS population, the effects of these biological mechanisms, along with socioeconomic and sociodemographic factors among individuals with diabetes, could be accelerating the cancer risk.

The main noticeable strength of our study is that two-thirds of our study participants represent the underserved and understudied populations. Our study also has a higher baseline prevalence of T2D (20%) coupled with a comprehensive assessment of potential demographic, lifestyle, and behavioral major disease risk factors, which allowed us to enable control for a wide range of potential confounders and carefully evaluate the association. However, a limitation of our study is that T2D was self-reported, and the specific date of T2D diagnosis is unclear, which could introduce potential exposure misclassification. Previous SCCS validation studies have shown that self-reported T2D is highly specific(23,28). We did not specifically ask for T2D diagnosis in the study, thus, some diabetes diagnoses could be type 1 diabetes. Given that type 1 diabetes prevalence is substantially lower than T2D in middle-aged to older populations, we could assume that the vast majority of diabetes diagnoses should be T2D. Some T2D patients may be undiagnosed or unreported in the population(28), which could also attenuate the true association between the exposure and outcomes. Some T2D patients may change their lifestyle after diagnosis, which could confound the association of prior T2D with the risk of certain cancers. Future studies with repeat surveys of lifestyle factors could be helpful to address this concern.

In conclusion, in this large prospective cohort study, we found that T2D was associated with the risk of overall cancer and certain site-specific cancers. Our findings were generally consistent with previous studies and provide strong support that T2D is a significant risk factor for cancer in low-income and minority populations. Our study stresses the importance of T2D prevention in reducing the incidence of these cancers.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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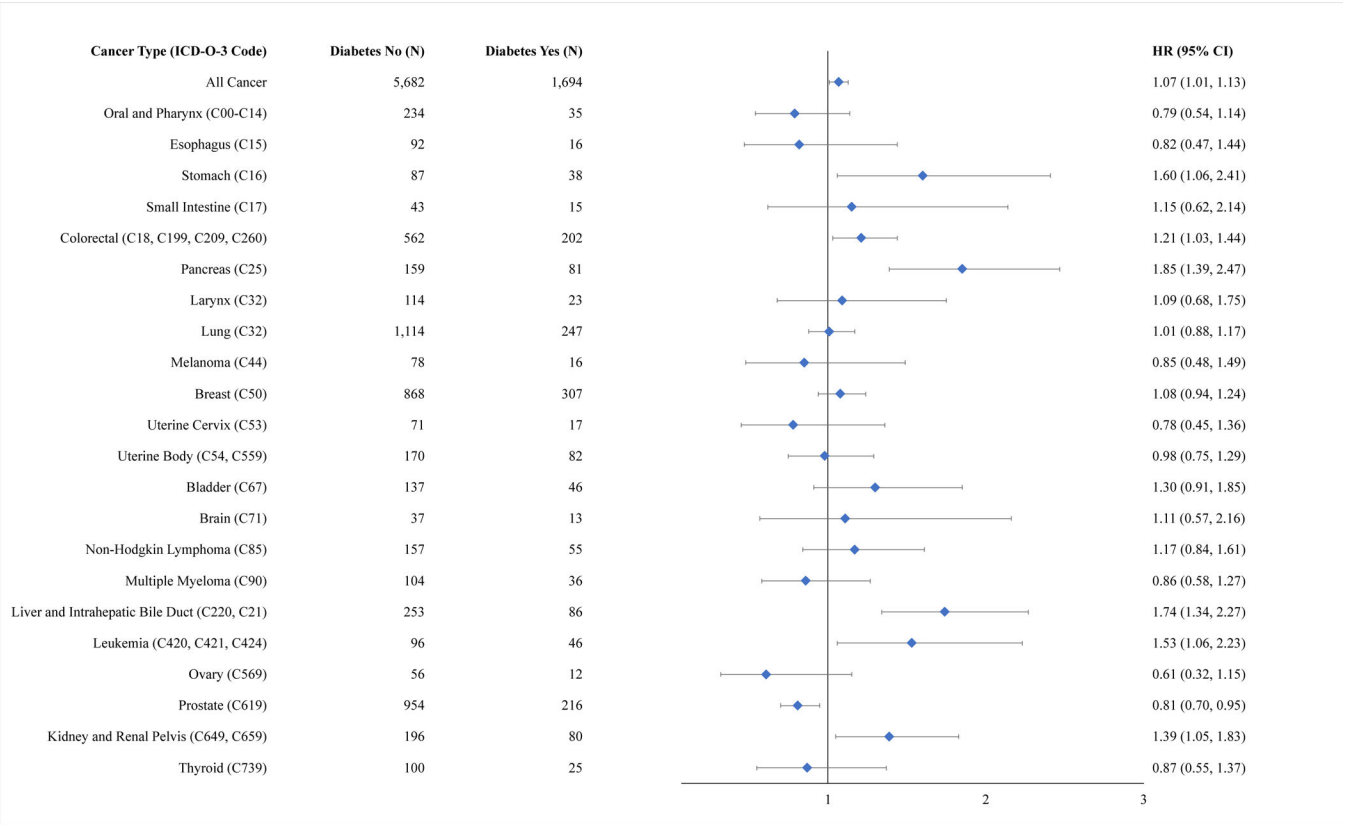


Figure 1: HRs and 95% CI of cancer associated with type 2 diabetes, Southern Community Cohort Study
Model adjusted for sex, race, education, income, enrollment source, smoking status, pack-years, and body-mass index.

Table 1.

Selected baseline characteristics of study participants by comorbidity, Southern Community Cohort Study

Baseline Characteristics	Total Cohort N = 76,121	Type 2 Diabetes	
		No N = 59,984	Yes N = 16,137
Age (Mean (SD), yr)	52.0 (8.67)	51.2 (8.46)	54.8 (8.87)
Race (N (%))			
White	22,044 (29.0%)	17,772 (29.6%)	4,272 (26.5%)
Black	51,121 (67.2%)	39,913 (66.5%)	11,208 (69.4%)
Other	2,956 (3.9%)	2,299 (3.8%)	657 (4.1%)
Sex (Male (N), %)	31,542 (41.4%)	25,792 (43.0%)	5,750 (35.6%)
Baseline body mass index (Mean (SD))	30.3 (7.56)	29.3 (7.12)	34.0 (8.04)
Education			
Less than high school	47,327 (62.2%)	36,582 (61.0%)	7,395 (45.8%)
Completed high school, GED, or other training	4,239 (5.6%)	3,358 (5.6%)	881 (5.5%)
Some college or junior college	14,904 (19.6%)	12,019 (20.0%)	2,885 (17.9%)
Graduated from college or further education	9,651 (12.7%)	8,025 (12.7%)	1,626 (10.1%)
Income (N (%))			
< \$15,000	42,362 (55.7%)	32,503 (54.2%)	9,859 (61.1%)
\$15,000 - \$24,999	16,217 (21.3%)	12,861 (21.4%)	3,356 (20.8%)
\$25,000 - \$49,999	10,573 (13.9%)	8,611 (14.4%)	1,962 (12.2%)
\$50,000 - \$99,999	5,229 (6.9%)	4,444 (7.4%)	785 (4.9%)
≥ \$100,000	1,740 (2.3%)	1,565 (2.6%)	175 (1.1%)
Smoking Status			
Current	31,584 (41.5%)	26,886 (44.8%)	4,698 (29.1%)
Former	17,104 (22.5%)	12,336 (20.6%)	4,768 (29.5%)
Never	27,433 (36.0%)	20,762 (34.6%)	6,671 (41.3%)
Packyears smoking (Mean (SD))	14.7 (21.2)	14.6 (20.5)	15.0 (23.6)

Table 2. HRs and 95% CI of cancer associated with type 2 diabetes by race, Southern Community Cohort Study

Ever Diagnosed with Diabetes	Among White			Among Black			P for heterogeneity ³
	Cases (n)	HR (95% CI) ¹	HR (95% CI) ²	Cases (n)	HR (95% CI) ¹	HR (95% CI) ²	
All Cancer							
No	1,670	1.00	1.00	3,857	1.00	1.00	
Yes	481	1.12 (1.01, 1.24)	1.18 (1.06, 1.31)	1,152	0.97 (0.90, 1.03)	1.02 (0.95, 1.09)	0.011
Oral and Pharynx							
No	63	1.00	1.00	169	1.00	1.00	
Yes	11	0.72 (0.38, 1.38)	0.86 (0.44, 1.71)	19	0.42 (0.26, 0.68)	0.60 (0.37, 0.98)	0.245
Esophagus							
No	19	1.00	1.00	73	1.00	1.00	
Yes	3	0.63 (0.18, 2.16)	0.65 (0.18, 2.41)	13	0.63 (0.35, 1.15)	0.84 (0.45, 1.57)	0.245
Stomach							
No	21	1.00	1.00	64	1.00	1.00	
Yes	8	1.39 (0.61, 3.17)	1.22 (0.51, 2.95)	30	1.44 (0.92, 2.24)	1.71 (1.07, 2.72)	0.919
Small Intestine							
No	10	1.00	1.00	33	1.00	1.00	
Yes	3	1.47 (0.39, 5.43)	1.34 (0.33, 5.41)	12	1.11 (0.56, 2.17)	1.07 (0.54, 2.13)	0.927
Colorectal							
No	134	1.00	1.00	411	1.00	1.00	
Yes	53	1.52 (1.10, 2.11)	1.62 (1.15, 2.29)	145	1.13 (0.93, 1.37)	1.13 (0.92, 1.37)	0.087
Pancreas							
No	45	1.00	1.00	109	1.00	1.00	
Yes	19	1.63 (0.94, 2.81)	1.58 (0.89, 2.82)	62	1.75 (1.27, 2.40)	2.05 (1.47, 2.86)	0.756
Larynx							
No	32	1.00	1.00	80	1.00	1.00	
Yes	4	0.45 (0.16, 1.29)	0.61 (0.20, 1.82)	18	0.88 (0.52, 1.49)	1.24 (0.72, 2.14)	0.276
Lung							
No	377	1.00	1.00	701	1.00	1.00	
Yes	89	0.79 (0.62, 0.99)	1.15 (0.90, 1.46)	144	0.65 (0.54, 0.78)	0.92 (0.76, 1.11)	0.044

Ever Diagnosed with Diabetes	Among White						Among Black			P for heterogeneity ³
	Cases (n)			HR (95% CI) ¹			HR (95% CI) ²			
	Cases (n)	HR (95% CI) ¹	HR (95% CI) ²	Cases (n)	HR (95% CI) ¹	HR (95% CI) ²	Cases (n)	HR (95% CI) ¹	HR (95% CI) ²	
Melanoma										
No	70	1.00	1.00				6	1.00	1.00	
Yes	15	0.92 (0.52, 1.62)	0.92 (0.51, 1.65)				1	(0.49 (0.06, 4.14)	0.41 (0.05, 3.60)	0.574
Breast										
No	266	1.00	1.00				579	1.00	1.00	
Yes	76	1.14 (0.88, 1.48)	1.04 (0.79, 1.36)				227	1.17 (1.00, 1.37)	1.12 (0.96, 1.32)	0.967
Uterine Cervix										
No	14	1.00	1.00				57	1.00	1.00	
Yes	6	1.80 (0.68, 4.76)	1.69 (0.60, 4.69)				9	0.48 (0.23, 0.97)	0.49 (0.24, 1.01)	0.060
Uterine Body										
No	47	1.00	1.00				120	1.00	1.00	
Yes	30	2.52 (1.57, 4.02)	1.55 (0.96, 2.51)				48	0.96 (0.68, 1.34)	0.74 (0.52, 1.04)	0.033
Bladder										
No	61	1.00	1.00				70	1.00	1.00	
Yes	16	0.98 (0.56, 1.70)	1.24 (0.69, 2.23)				25	1.11 (0.70, 1.77)	1.16 (0.72, 1.88)	0.632
Brain										
No	15	1.00	1.00				21	1.00	1.00	
Yes	7	1.75 (0.71, 4.36)	1.85 (0.70, 4.86)				6	0.90 (0.36, 2.28)	0.82 (0.32, 2.11)	0.360
Non-Hodgkin Lymphoma										
No	59	1.00	1.00				92	1.00	1.00	
Yes	20	1.33 (0.79, 2.23)	1.13 (0.66, 1.94)				34	1.30 (0.87, 1.95)	1.25 (0.82, 1.90)	0.848
Multiple Myeloma										
No	23	1.00	1.00				81	1.00	1.00	
Yes	5	0.84 (0.31, 2.23)	0.60 (0.22, 1.66)				31	1.05 (0.69, 1.60)	0.90 (0.59, 1.39)	0.626
Liver and Intrahepatic Bile Duct										
No	44	1.00	1.00				203	1.00	1.00	
Yes	22	2.00 (0.19, 3.38)	2.94 (1.66, 5.20)				58	1.10 (0.82, 1.49)	1.43 (1.05, 1.95)	0.049
Leukemia										
No	45	1.00	1.00				47	1.00	1.00	
Yes	15	1.38 (0.76, 2.51)	1.44 (0.77, 2.67)				29	1.81 (1.12, 2.90)	1.65 (1.01, 2.69)	0.329

Ever Diagnosed with Diabetes	Among White			Among Black			<i>P</i> for heterogeneity ³
	Cases (n)	HR (95% CI) ¹	HR (95% CI) ²	Cases (n)	HR (95% CI) ¹	HR (95% CI) ²	
Ovary							
No	20	1.00	1.00	35	1.00	1.00	
Yes	2	0.36 (0.08, 1.54)	0.40 (0.09, 1.78)	9	0.64 (0.30, 1.34)	0.62 (0.29, 1.32)	0.435
Prostate							
No	206	1.00	1.00	725	1.00	1.00	
Yes	44	0.98 (0.70, 1.36)	1.04 (0.74, 1.46)	163	0.83 (0.70, 0.98)	0.77 (0.64, 0.92)	0.420
Kidney and Renal Pelvis							
No	61	1.00	1.00	123	1.00	1.00	
Yes	24	1.61 (0.99, 2.59)	1.36 (0.82, 2.24)	53	1.46 (1.05, 2.03)	1.43 (1.02, 2.01)	0.930
Thyroid							
No	38	1.00	1.00	58	1.00	1.00	
Yes	9	1.03 (0.50, 2.16)	0.94 (0.43, 2.01)	16	0.98 (0.56, 1.72)	0.86 (0.48, 1.52)	0.936

¹Model adjusted for sex, education, income, and enrollment source.

²Model adjusted for sex, education, income, enrollment source, smoking status, pack-years, and body mass index.

³*P* for heterogeneity calculated for model 2 adjustments.

Table 3.

HRs and 95% CI of cancer associated with diabetes by years between diagnosis of type 2 diabetes and baseline survey, Southern Community Cohort Study

Cancer Type and Diabetes Status	Total Cases (n) ²	<10 years	10–19.99 years	≥ 20 years
		HR (95% CI) ¹	HR (95% CI) ¹	HR (95% CI) ¹
All Cancer				
No	5,682	1.00	1.00	
Yes	1,694	1.10 (1.03, 1.18)	1.02 (0.92, 1.13)	0.96 (0.83, 1.09)
Colorectal				
No	562	1.00	1.00	
Yes	202	1.27 (1.04, 1.55)	0.96 (0.69, 1.34)	1.31 (0.91, 1.87)
Pancreas				
No	159	1.00	1.00	
Yes	81	1.99 (1.43, 2.75)	1.30 (0.74, 2.27)	2.01 (1.14, 3.52)
Lung				
No	1,114	1.00	1.00	
Yes	247	1.04 (0.87, 1.24)	0.96 (0.73, 1.26)	0.96 (0.68, 1.34)
Breast				
No	868	1.00	1.00	
Yes	307	1.06 (0.90, 1.25)	1.15 (0.90, 1.45)	1.02 (0.75, 1.39)
Uterine Body				
No	170	1.00	1.00	
Yes	82	1.14 (0.84, 1.56)	0.54 (0.29, 0.99)	1.17 (0.68, 2.01)
Non-Hodgkin Lymphoma				
No	157	1.00	1.00	
Yes	55	1.28 (0.88, 1.65)	1.12 (0.63, 2.00)	0.59 (0.22, 1.62)
Liver and Intrahepatic Bile Duct				
No	253	1.00	1.00	
Yes	86	1.92 (1.42, 2.58)	1.83 (1.14, 2.92)	1.07 (0.50, 2.29)
Prostate				
No	954	1.00	1.00	
Yes	216	0.83 (0.69, 1.00)	0.75 (0.56, 0.99)	0.66 (0.44, 1.00)
Kidney and Renal Pelvis				
No	196	1.00	1.00	
Yes	80	1.49 (1.08, 2.04)	1.52 (0.95, 2.44)	0.96 (0.45, 2.06)

¹Model adjusted for sex, race, education, income, enrollment source, smoking status, pack-years, and body mass index.

²Reference group is individuals without diabetes.

Table 4. HRs and 95% CI of cancer associated with diabetes by years since diabetes diagnosis, Southern Community Cohort Study

Cancer Type and Diabetes Status	Total Cases (n) ²	<15 years		15–29.99 years		≥30 years	
		HR (95% CI) ¹	HR (95% CI) ¹	HR (95% CI) ¹	HR (95% CI) ¹	HR (95% CI) ¹	HR (95% CI) ¹
All Cancer							
No	5,682	1.00	1.00	1.00	1.00	1.00	1.00
Yes	1,694	1.10 (1.02, 1.17)	1.04 (0.95, 1.14)	0.88((0.73, 1.05)			
Colorectal							
No	562	1.00	1.00	1.00	1.00	1.00	1.00
Yes	202	1.20 (0.98, 1.48)	1.19 (0.91, 1.55)	1.22 (0.77, 1.94)			
Pancreas							
No	159	1.00	1.00	1.00	1.00	1.00	1.00
Yes	81	1.92 (1.37, 2.69)	1.84 (1.21, 2.82)	1.32 (0.58, 3.01)			
Lung							
No	1,114	1.00	1.00	1.00	1.00	1.00	1.00
Yes	247	1.08 (0.91, 1.29)	0.84 (0.65, 1.09)	1.02 (0.68, 1.53)			
Breast							
No	868	1.00	1.00	1.00	1.00	1.00	1.00
Yes	307	1.03 (0.87, 1.22)	1.19 (0.97, 1.47)	0.96 (0.65, 1.43)			
Uterine Body							
No	170	1.00	1.00	1.00	1.00	1.00	1.00
Yes	82	1.22 (0.89, 1.66)	0.56 (0.33, 0.95)	1.16 (0.61, 2.21)			
Non-Hodgkin Lymphoma							
No	157	1.00	1.00	1.00	1.00	1.00	1.00
Yes	55	1.20 (0.82, 1.75)	1.28 (0.78, 2.10)	0.24 (0.03, 1.74)			
Liver and Intrahepatic Bile Duct							
No	253	1.00	1.00	1.00	1.00	1.00	1.00
Yes	86	1.95 (1.44, 2.66)	1.66 (1.09, 2.51)	1.18 (0.48, 2.87)			
Prostate							
No	954	1.00	1.00	1.00	1.00	1.00	1.00
Yes	216	0.86 (0.71, 1.03)	0.70 (0.54, 0.91)	0.59 (0.33, 1.04)			

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Cancer Type and Diabetes Status	Total Cases (n) ²	<15 years		15–29.99 years		≥30 years	
		HR (95% CI) ¹		HR (95% CI) ¹		HR (95% CI) ¹	
Kidney and Renal Pelvis							
No	196	1.00		1.00		1.00	
Yes	80	1.57 (1.14, 2.16)		1.43 (0.93, 2.18)		0.42 (0.10, 1.70)	

¹Model adjusted for sex, race, education, income, enrollment source, smoking status, pack-years, and body mass index.
²Reference group is individuals without diabetes.